

Unit 6&7 - PM

MC – Questions
Answered on D2L

MC	Written	Total
/14	/21	/35

Knowledge

1. Solve. Check for extraneous roots. Round to the nearest hundredth where necessary.

a) $8^{x-6}(128^{2x}) + 15 = 110$

(1.5)

b) $\log_a(ab) - \log_a(a^3b)$

(1.5)

2. Graph the function $f(x) = 2\log_4(2x + 10) - 1$ by using the mapping rule and solve for the key features algebraically. Label the graph. Leave solutions to two decimal places if necessary. (5 marks)

(graph: 1)

a) domain _____(0.5)

b) equation of the vertical asymptote _____(0.5)

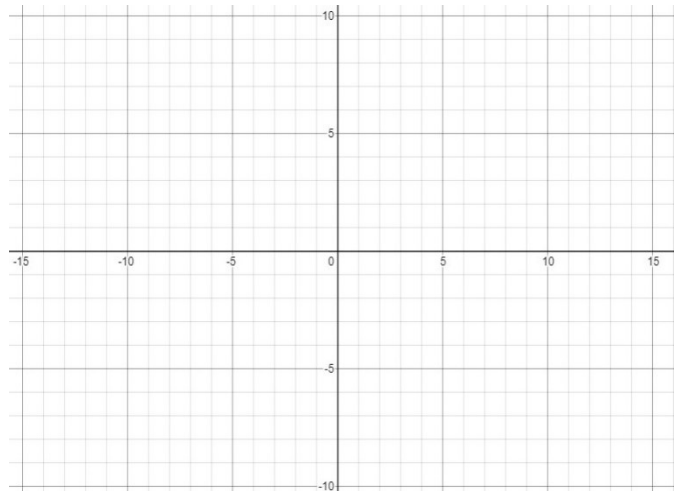
c) mapping rule _____(1)

d) x-intercept

(1)

e) y-intercept:

(1)



COMMUNICATION

1. Describe the transformations you would apply to $f(x) = \log x$ to obtain $g(x) = -2\log(4 - 2x)$

(2)

APPLICATION

1. Evaluate. Check for extraneous roots.

a) $\log_3 \sqrt{x^2 - 3x} = \log_3 2$

(2)

b) $\log_3(x + 4) + \log_3(2x - 3) = \log_3(6)$

(3)

2. Tom wants to buy his dream car that sells for \$76 000. He only has half of that amount and invests it at 5.5%/a, compounded monthly, for five years. After the five years, he is still short. How much longer will he have to invest his money in another account earning interest at 8%/a, compounded daily, to have enough for his dream car? Provide your answer to the approximate year.

(3)

THINKING

1. If $\left(\frac{1}{2}\right)^{x+y} = 16$ and $\log_{x-y} 8 = -3$, calculate the values of x and y . Exact solutions only.

(3)